

SHIMENG YU

professor

基本信息

姓名: SHIMENG YU

所属院校: 佐治亚理工学院

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地区: United States

学科领域

集成电路设计与集成系统

微电子科学与工程

人工智能

电子与计算机工程

研究领域

存储器设计与架构

微电子器件与制造

深度学习与神经网络

纳米电子与先进工艺

计算机体系结构

低功耗电路与能量管理

高性能计算芯片

低功耗设计与能耗优化

集成电路设计与制造

个人简介

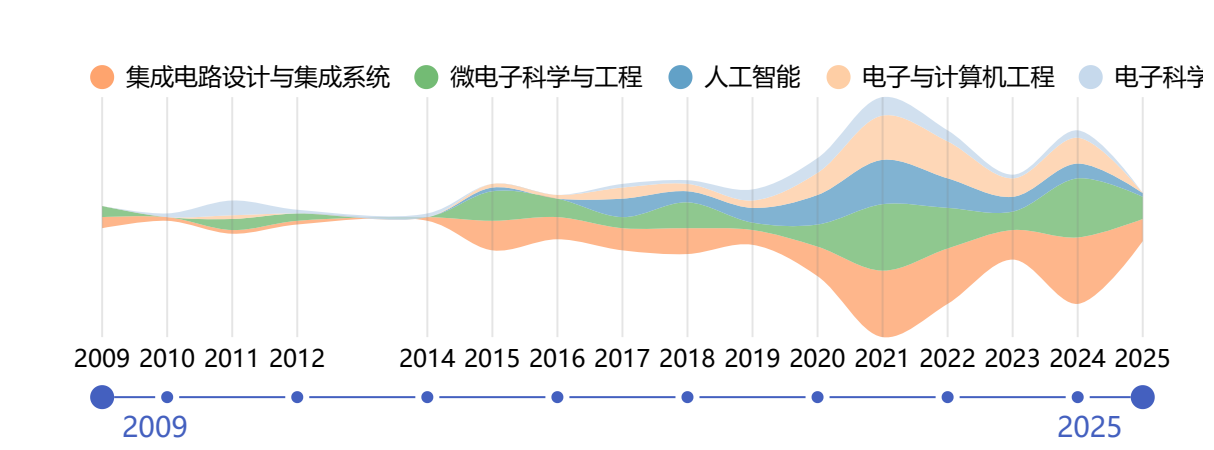
Shimeng Yu is a professor at the School of Electrical and Computer Engineering at the Georgia Institute of Technology. He earned his B.S. in microelectronics from Peking University and his M.S. and Ph.D. in electrical engineering from Stanford University. His research focuses on semiconductor devices and integrated circuits for energy-efficient computing systems, particularly non-volatile memories for AI hardware and in-memory computing. Yu has received numerous awards, including the NSF Faculty Early CAREER Award and IEEE EDS Early Career Award. He is an active member of professional societies and serves on the technical program committees of several leading conferences.

工作经历

教育经历

Peking University - B.S.	- 2009
Stanford University - M.S.	- 2011
Stanford University - Ph.D.	- 2013

近年发表的论文



Design of Superlattice Ferroelectric-Metal Field-effect Transistor for triple-level cell 3D NAND flash
MICROELECTRONIC ENGINEERING
Woo, Sola; Choe, Gihun; Khan, Asif Islam; Datta, Suman; Yu, Shimeng
被引用次数: 0 2025

Signal Margin, Density, and Scalability of 3-D DRAM: A Comparative Study of Two Bitline Architectures
IEEE TRANSACTIONS ON ELECTRON DEVICES
Wu, Xiangjin; Upton, Luke R.; Chen, Jian; Hsu, Po-Kai; Yu, Shimeng; Wong, H. -S. Philip
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Effect of DC Stress on Low-Frequency Noise Characteristics of W-Doped In2O3 BEOL Transistors
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Phadke, Omkar; Aabrar, Khandker Akif; Choe, Gihun; Luo, Yuan-Chun; Kirtania, Sharadindu Gopal; Khan, Asif Islam; Datta, Suman; Yu, Shimeng
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Optimizing Memory Window for Ferroelectric Nand Applications: An Experimental Study on Dielectric Material Selection and Layer Positioning
IEEE TRANSACTIONS ON ELECTRON DEVICES

Fernandes, Lance; Ravindran, Prasanna Venkatesan; Chen, Jiayi; Tian, Mengkun; Das, Dipjyoti; Chen, Hang; Chern, Winston; Kim, Kijoon; Woo, Jongho; Lim, Suhwan; Kim, Kwangsoo; Kim, Wanki; Ha, Daewon; Yu, Shimeng; Datta, Suman; Khan, Asif

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2025

Capacitive Crossbar Array for Solving Matrix Equations in One-Shot

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Vadlamani, Madhav; Jia, Jianwei; Xie, Tian; Luo, Yuan-Chun; Lee, Junmo; Li, Shaolan; Yu, Shimeng

被引用次数: 0

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Demonstration of Robust Retention in Band Engineered FEFETs for NAND Storage Applications Using Tunnel Dielectric Layer

IEEE ELECTRON DEVICE LETTERS

Venkatesan, Prasanna; Fernandes, Lance; Ravikumar, Priyanka; Park, Chinsung; Tran, Huy; Wang, Zekai; Jayasankar, Hari; Garlapati, Amrit; Song, Taeyoung; Tian, Mengkun; Chen, Hang; Chern, Winston; Kim, Kijoon; Woo, Jongho; Lim, Suhwan; Kim, Kwangsoo; Yu, Shimeng; Datta, Suman; Kim, Wanki; Ha, Daewon; Khan, Asif

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Opposing Mean Error Compensation for Accuracy Enhancement in Analog Compute-in-Memory With Resistive Switching Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Huang, Wei-Hsing; Lee, Woocheol; Kim, Joong Sik; Koo, Won-Tae; Suh, Dong Ik; Lee, Seho; Yi, Jaeyun; Cha, Seon Yong; Yu, Shimeng

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2025

The origin of memory window closure with bipolar stress cycling in silicon ferroelectric field-effect-transistors

JOURNAL OF APPLIED PHYSICS

Passlack, Matthias; Tasneem, Nujhat; Park, Chinsung; Ravindran, Prasanna Venkat; Chen, Hang; Das, Dipjyoti; Yu, Shimeng; Chen, Edward; Wang, Jer-Fu; Chang, Chih-Sheng; Lin, Yu-Ming; Radu, Iuliana; Khan, Asif

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Paving the Way for Pass Disturb-Free Vertical NAND Storage via a Dedicated and String-Compatible Pass Gate

ACS APPLIED MATERIALS & INTERFACES

Zhao, Zijian; Woo, Sola; Aabrar, Khandker Akif; Kirtania, Sharadindu Gopal; Jiang, Zhouhang; Deng, Shan; Xiao, Yi; Mulaosmanovic, Halid; Duenkel, Stefan; Kleimaier, Dominik; Soss, Steven; Beyer, Sven; Joshi, Rajiv; Meninger, Scott; Mohamed, Mohamed; Kim, Kijoon; Woo, Jongho; Lim, Suhwan; Kim, Kwangsoo; Kim, Wanki; Ha, Daewon; Narayanan, Vijaykrishnan; Datta, Suman; Yu, Shimeng; Ni, Kai

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A FeFET-Based ADC Offset Robust Compute-In-Memory Architecture for Streaming Keyword Spotting (KWS)

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H3D-Transformer: A Heterogeneous 3D (H3D) Computing Platform for Transformer Model Acceleration on Edge Devices

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2024

Semiconductor Memory Technologies: State-of-the-Art and Future Trends

COMPUTER

Yu, Shimeng; Kim, Tae-Hyeon

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A Heterogeneous Platform for 3D NAND-Based In-Memory Hyperdimensional Computing Engine for Genome Sequencing Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

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Material Choices for Tunnel Dielectric Layer and Gate Blocking Layer for Ferroelectric NAND Applications

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Capacitive Synaptor With Overturned Charge Injection for Compute-in-Memory

IEEE ELECTRON DEVICE LETTERS

Kim, Choong-Ki; Phadke, Omkar; Kim, Tae-Hyeon; Kim, Myung-Su; Yu, Ji-Man; Yoo, Min-Soo; Choi, Yang-Kyu; Yu, Shimeng

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2024

Suppressed Capacitive Coupling in 2 Transistor Gain Cell With Oxide Channel and Split Gate

IEEE TRANSACTIONS ON ELECTRON DEVICES

Phadke, Omkar; Kirtania, Sharadindu Gopal; Chakraborty, Dyutimoy; Datta, Suman; Yu, Shimeng

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Thermally Constrained Codesign of Heterogeneous 3-D Integration of Compute-in-Memory, Digital ML Accelerator, and RISC-V Cores for Mixed ML and Non-ML Workloads

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Luo, Yuan-Chun; Lu, Anni; Sharda, Janak; Scherer, Moritz; Gomez, Jorge Tomas; Sarwar, Syed Shakib; Li, Ziyun; Pinkham, Reid Frederick; Salvo, Barbara De; Yu, Shimeng

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IEEE TRANSACTIONS ON ELECTRON DEVICES

Luo, Yuan-Chun; Khanna, Abhishek; Grisafe, Benjamin; Sun, Jiaxin; Dutta, Sourav; Noskin, Lindsey E.; Adamo, Carolina; Mei, Antonio B.; Ghosh, Ram Krishna; Colletta, Michael; Holtz, M. E.; Gambin, Vincent; Kourkoutis, Lena F.; Yu, Shimeng; Schlom, Darrell G.; Datta, Suman

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Machine Learning-Assisted Compact Modeling of W-Doped Indium Oxide Channel Transistor for Back-End-of-Line Applications

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Analyzing Total-Ionizing-Dose Induced Memory Window Degradation in Ferroelectric FinFET

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Design-Technology Co-Optimization for Stacked Nanosheet Oxide Channel Transistors in Monolithic 3D Integrated Circuit Design

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Kwak, Jungyoung; Choe, Gihun; Yu, Shimeng

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HyperGen: compact and efficient genome sketching using hyperdimensional vectors

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Xu, Weihong; Hsu, Po-Kai; Moshiri, Niema; Yu, Shimeng; Rosing, Tajana

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Jongho; Lim, Suhwan; Kim, Kwangsoo; Kim, Wanki; Ha, Daewon; Mahapatra, Souvik; Yu, Shimeng; Datta, Suman; Khan, Asif

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BEOL-Compatible On-Chip DC-DC Converters

2024 IEEE INTERNATIONAL INTERCONNECT TECHNOLOGY CONFERENCE, IITC 2024

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3D Digital Compute-in-Memory Benchmark with A5 CFET Technology

2024 IEEE THE 20TH ASIA PACIFIC CONFERENCE ON CIRCUITS AND SYSTEMS, APCCAS 2024

Shon, Minji; Lee, Junmo; Yu, Shimeng

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Modeling Row Hammer Effect in 3D Capacitor-less DRAM using Triple-Gated Silicon Nanosheet Device

2024 INTERNATIONAL CONFERENCE ON SIMULATION OF SEMICONDUCTOR PROCESSES AND DEVICES, SISPAD 2024

Son, Jimin; Park, Jun Young; Lee, Taeun; Woo, Sola; Yu, Shimeng

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A Comparative Study of HBL and VBL 3D DRAM: Signal Margin, Bit-cell Density, and Scalability

2024 IEEE SILICON NANOELECTRONICS WORKSHOP, SNW 2024

Wu, Xiangjin; Upton, Luke R.; Hsu, Po-Kai; Chen, Jian; Yu, Shimeng; Wong, H-S Philip

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A Reconfigurable Bandpass Filter with Ferroelectric Devices for Intracardiac Electrograms Monitoring

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Jia, Jianwei; Jia, Zhenge; Phadke, Omkar; Choe, Gihun; Shi, Yiyu; Yu, Shimeng

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Impact of In-Pixel Processing Circuit Non-idealities on Multi-object Tracking Accuracy for Autonomous Driving

2024 IEEE 67TH INTERNATIONAL MIDWEST SYMPOSIUM ON CIRCUITS AND SYSTEMS, MWSCAS 2024

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A Dynamic Power-Only Compute-in-Memory Macro With Power-of-Two Nonlinear SAR ADC for Nonvolatile Ferroelectric Capacitive Crossbar Array

IEEE SOLID-STATE CIRCUITS LETTERS

Yeo, Injune; He, Wangxin; Luo, Yuan-Chun; Yu, Shimeng; Seo, Jae-Sun

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Design Framework for Ferroelectric Gate Stack Engineering of Vertical NAND Structures for Efficient TLC and QLC Operation

2024 IEEE INTERNATIONAL MEMORY WORKSHOP, IMW

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2024

Engineering nvCap From FEOL to BEOL with Ferroelectric Small-signal Non-destructive Read

2024 IEEE INTERNATIONAL MEMORY WORKSHOP, IMW

Kim, Tae-Hyeon; Luo, Yuan-Chun; Phadke, Omkar; Read, James; Yu, Shimeng

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Monolithic 3D Transposable 3T Embedded DRAM with Back-end-of-line Oxide Channel Transistor

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Kwak, Jungyouon; Choe, Gihun; Lee, Junmo; Yu, Shimeng

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2024

A Cross-layer Framework for Design Space and Variation Analysis of Non-Volatile Ferroelectric Capacitor-Based Compute-in-Memory Accelerators

29TH ASIA AND SOUTH PACIFIC DESIGN AUTOMATION CONFERENCE, ASP-DAC 2024

Luo, Yuan-Chun; Read, James; Lu, Anni; Yu, Shimeng

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Reliability Assesement of Ferroelectric nvCAP for Small-signal Capacitive Read-out

2024 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM, IRPS 2024

Phadke, Omkar; Mulaosmanovic, Halid; Duenkel, Stefan; Beyer, Sven; Yu, Shimeng

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Comprehensive time dependent dielectric breakdown (TDDB) characterization of ferroelectric capacitors under bipolar stress conditions

2024 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM, IRPS 2024

Ravikumar, Priyanka Gundlapudi; Ravindran, Prasanna Venkatesan; Aabrar, Khandker Akif; Song, Taeyoung; Kirtania, Sharadindu Gopal; Das, Dipjyoti; Park, Chinsung; Afroze, Nashrah; Tian, Mengkun; Yu, Shimeng; Islam, Ahmad Ehtesham; Datta, Suman; Mahapatra, Souvik; Khan, Asif

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2024

Accelerator Design using 3D Stacked Capacitorless DRAM for Large Language Models

2024 IEEE 6TH INTERNATIONAL CONFERENCE ON AI CIRCUITS AND SYSTEMS, AICAS 2024

Sharda, Janak; Hsu, Po-Kai; Yu, Shimeng

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2024

Digital CIM with Noisy SRAM Bit: A Compact Clustered Annealer for Large-Scale Combinatorial Optimization

Design Automation Conference

Anni Lu; Junmo Lee; Yuan-Chun Luo; Hai Li; Ian Young; Shimeng Yu

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2024

Hardware-aware Quantization/Mapping Strategies for Compute-in-Memory Accelerators

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Huang, Shanshi; Jiang, Hongwu; Yu, Shimeng

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Temporal Frame Filtering for Autonomous Driving Using 3D-Stacked Global Shutter CIS With IWO Buffer Memory and Near-Pixel Compute

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

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ENNA: An Efficient Neural Network Accelerator Design Based on ADC-Free Compute-In-Memory Subarrays

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Jiang, Hongwu; Huang, Shanshi; Li, Wantong; Yu, Shimeng

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Tunable Non-Volatile Gate-to-Source/Drain Capacitance of FeFET for Capacitive Synapse

IEEE ELECTRON DEVICE LETTERS

Kim, Tae-Hyeon; Phadke, Omkar; Luo, Yuan-Chun; Mulaosmanovic, Halid; Mueller, Johannes; Duenkel, Stefan; Beyer, Sven; Khan, Asif Islam; Datta, Suman; Yu, Shimeng

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2023

Capacitive Memory Window With Non-Destructive Read in Ferroelectric Capacitors

IEEE ELECTRON DEVICE LETTERS

Mukherjee, Shankha; Bizindavyi, Jasper; Clima, Sergiu; Popovici, Mihaela I.; Piao, Xiaoyu; Katcko, Kostantine; Catthoor, Francky; Yu, Shimeng; Afanas'ev, Valeri V.; Van Houdt, Jan

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2023

A Ge-Channel Ferroelectric Field Effect Transistor With Logic-Compatible Write Voltage

IEEE ELECTRON DEVICE LETTERS

Das, Dipjyoti; Ravindran, Prasanna Venkatesan; Park, Chinsung; Tasneem, Nujhat; Wang, Zheng; Chen, Hang; Chern, Winston; Yu, Shimeng; Datta, Suman; Khan, Asif

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2023

Scalable In-Memory Clustered Annealer With Temporal Noise of Charge Trap Transistor for Large Scale Travelling Salesman Problems

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Lu, Anni; Hur, Jae; Luo, Yuan-Chun; Li, Hai; Nikonov, Dmitri E.; Young, Ian A.; Choi, Yang-Kyu; Yu, Shimeng

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2023

H3DAtten: Heterogeneous 3-D Integrated Hybrid Analog and Digital Compute-in-Memory Accelerator for Vision Transformer Self-Attention

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Li, Wantong; Manley, Madison; Read, James; Kaul, Ankit; Bakir, Muhannad S.; Yu, Shimeng

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Interfacial Oxide Layer Scavenging in Ferroelectric Hf_{0.5}Zr_{0.5}O₂-Based MOS Structures With Ge Channel for Reduced Write Voltages

IEEE TRANSACTIONS ON ELECTRON DEVICES

Park, Chinsung; Kashyap, Harshil; Das, Dipjyoti; Hur, Jae; Tasneem, Nujhat; Lombardo, Sarah; Afroze, Nashrah; Chern, Winston; Kummel, Andrew C.; Yu, Shimeng; Khan, Asif Islam

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2023

Machine Learning-Assisted Statistical Variation Analysis of Ferroelectric Transistor: From Experimental Metrology to Adaptive Modeling

IEEE TRANSACTIONS ON ELECTRON DEVICES

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3-D Heterogeneous Integration of RRAM-Based Compute-In-Memory: Impact of Integration Parameters on Inference Accuracy

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kaul, Ankit; Luo, Yandong; Peng, Xiaochen; Manley, Madison; Luo, Yuan-Chun; Yu, Shimeng; Bakir, Muhannad S.

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2023

Immediate Read-After-Write Capability in p-Type Ferroelectric Field-Effect Transistors and Its Evolution With Fatigue Cycling

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Tasneem, Nujhat; Wang, Zheng; Chen, Hang; Yu, Shimeng; Chern, Winston; Khan, Asif

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2023

A High-Parallelism RRAM-Based Compute-In-Memory Macro With Intrinsic Impedance Boosting and In-ADC Computing

IEEE JOURNAL ON EXPLORATORY SOLID-STATE COMPUTATIONAL DEVICES AND CIRCUITS

Xie, Tian; Yu, Shimeng; Li, Shaolan

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An In-Storage Processing Architecture with 3D NAND Heterogeneous Integration for Spectra Open Modification Search

PROCEEDINGS OF THE INTERNATIONAL SYMPOSIUM ON MEMORY SYSTEMS, MEMSYS 2023

Hsu, Po-Kai; Xu, Weihong; Rosing, Tajana; Yu, Shimeng

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PRIVE: Efficient RRAM Programming with Chip Verification for RRAM-based In-Memory Computing Acceleration

2023 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

He, Wangxin; Meng, Jian; Gonugondla, Sujana Kumar; Yu, Shimeng; Shanbhag, Naresh R.; Seo, Jae-sun

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RAWatten: Reconfigurable Accelerator for Window Attention in Hierarchical Vision Transformers

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Li, Wantong; Luo, Yandong; Yu, Shimeng

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2023

Editorial: Novel materials, devices and solutions for brain-inspired sensing and computing
FRONTIERS IN NEUROSCIENCE

Monzio Compagnoni, Christian; Yu, Shimeng; Zhao, Weisheng; Ielmini, Daniele

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2022

Two-Way Transpose Multibit 6T SRAM Computing-in-Memory Macro for Inference-Training AI Edge Chips

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Su, Jian-Wei; Si, Xin; Chou, Yen-Chi; Chang, Ting-Wei; Huang, Wei-Hsing; Tu, Yung-Ning; Liu, Ruhui; Lu, Pei-Jung; Liu, Ta-Wei; Wang, Jing-Hong; Chung, Yen-Lin; Ren, Jin-Sheng; Chang, Fu-Chun; Wu, Yuan; Jiang, Hongwu; Huang, Shanshi; Li, Sih-Han; Sheu, Shyh-Shyuan; Wu, Chih-, I; Lo, Chung-Chuan; Liu, Ren-Shuo; Hsieh, Chih-Cheng; Tang, Kea-Tiong; Yu, Shimeng; Chang, Meng-Fan

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Remote Oxygen Scavenging of the Interfacial Oxide Layer in Ferroelectric Hafnium-Zirconium Oxide-Based Metal-Oxide-Semiconductor Structures

ACS APPLIED MATERIALS & INTERFACES

Tasneem, Nujhat; Kashyap, Harshil; Chae, Kisung; Park, Chinsung; Lee, Ping-che; Lombardo, Sarah F.; Afroze, Nashrah; Tian, Mengkun; Kumarasubramanian, Harish; Hur, Jae; Chen, Hang; Chern, Winston; Yu, Shimeng; Bandaru, Prabhakar; Ravichandran, Jayakanth; Cho, Kyeongjae; Kacher, Josh; Kummel, Andrew C.; Khan, Asif Islam

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A Ferroelectric-Based Volatile/Non-Volatile Dual-Mode Buffer Memory for Deep Neural Network Accelerators

IEEE TRANSACTIONS ON COMPUTERS

Luo, Yandong; Luo, Yuan-Chun; Yu, Shimeng

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Robust Processing-In-Memory With Multibit ReRAM Using Hessian-Driven Mixed-Precision Computation

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Dash, Saurabh; Luo, Yandong; Lu, Anni; Yu, Shimeng; Mukhopadhyay, Saibal

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Temperature-Resilient RRAM-Based In-Memory Computing for DNN Inference

IEEE MICRO

Meng, Jian; Shim, Wonbo; Yang, Li; Yeo, Injune; Fan, Deliang; Yu, Shimeng; Seo, Jae-sun

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Nonvolatile Capacitive Crossbar Array for In-Memory Computing

ADVANCED INTELLIGENT SYSTEMS

Hur, Jae; Luo, Yuan-Chun; Lu, Anni; Wang, Tzu-Han; Li, Shaolan; Khan, Asif Islam; Yu, Shimeng

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Accelerating On-Chip Training with Ferroelectric-Based Hybrid Precision Synapse

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

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Achieving High In Situ Training Accuracy and Energy Efficiency with Analog Non-Volatile Synaptic Devices

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Huang, Shanshi; Sun, Xiaoyu; Peng, Xiaochen; Jiang, Hongwu; Yu, Shimeng

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A 40-nm MLC-RRAM Compute-in-Memory Macro With Sparsity Control, On-Chip Write-Verify, and Temperature-Independent ADC References

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Li, Wantong; Sun, Xiaoyu; Huang, Shanshi; Jiang, Hongwu; Yu, Shimeng

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Design and Optimization of Non-Volatile Capacitive Crossbar Array for In-Memory Computing

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Luo, Yuan-Chun; Lu, Anni; Hur, Jae; Li, Shaolan; Yu, Shimeng

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2022

3D AND-Type Ferroelectric Transistors for Compute-in-Memory and the Variability Analysis

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MAC-ECC: In-Situ Error Correction and Its Design Methodology for Reliable NVM-Based Compute-in-Memory Inference Engine

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Li, Wantong; Read, James; Jiang, Hongwu; Yu, Shimeng

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2022

Guest Editorial Memristive Circuits and Systems for Edge-Computing Applications

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Sirakoulis, Georgios Ch.; Ascoli, Alon; Tetzlaff, Ronald; Yu, Shimeng

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2022

A Compute-in-Memory Hardware Accelerator Design With Back-End-of-Line (BEOL) Transistor Based Reconfigurable Interconnect

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

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GP3D: 3D NAND Based In-Memory Graph Processing Accelerator

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

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Characterizing HfO₂-Based Ferroelectric Tunnel Junction in Cryogenic Temperature

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